Network Analysis on Rumination

2. Methods

2.1 Network analysis

Psychological networks have their own characteristics which differentiate them from other networks of the graph theory (Epskamp, Borsboom, & Fried, 2017; Newman, 2010). Nodes usually represent some psychological measures or items of a psychological assessment tool, the edges of the network reflect the corresponding estimates of association derived from the data. Sample sizes have direct impact on estimated accuracy and stability of the psychological network, thus the accuracy of the network parameters and measures is crucial: By obtaining robust results we can have a more realistic picture of a network without knowing the true network shape and structure. Just like any other psychological variables, network characteristics are also sensitive to sampling, estimation methods, included variables, outliers and various other components of an analysis. As (Epskamp, Borsboom, Fried, et al., 2017) points out, psychological networks differ from *reflective latent variable models* by not seeking for a common cause of co-occurrence of variables (such as symptoms or different items of a questionnaire), but instead their focus is on quantifying and visually show the multivariate dependencies and the different pathways of direct relationships between the nodes. In the current study, we used a fairly large sample as shown above. Before going further into network modeling, we estimated the 95% confidence intervals of network centrality measures to have an understanding of the sampling distribution of our estimates. According to Epskamp, Borsboom, Fried, et al. (2017) low sample size and non-regularized estimates might lead to amplify any bootstrap bias; by having a larger sample of individuals, we expect lesser levels of bias of any bootstrap estimates for network accuracy and stability.

2.1.1 Removing redundant items As a preparation step, we applied the *goldbricker* function from the R package *networktools* to search for pairs of items in the RRS, SCRS and MHS items which were highly correlated ($r > 0.50$) and showed highly the same pattern with other items. When such "bad pairs" were identified, reduction techniques were put into use (with the *net_reduce* function in R): instead of the pair, the first principal component of the two variables were kept in the data as a new, merged variable.

The RRS items required no reduction steps as the highly correlating items did not extend their pattern to more nodes (their correlations to other nodes did not violate the criteria that > 75% of these correlations did not differ significantly).

2.1.2 Network Analysis with LASSO For the network analysis, the tuning parameter of *gamma* in the grahical LASSO alogrithm has been set to 0.5 value in accordance with (Bernstein et al., 2019) in order to achieve sparse graphs. The final networks were chosen by the lowest values of the *extended Bayesian Information Criterion* (EBIC).

2.1.3. Community detection In order to get an understanding of community of nodes and to see if nodes can be grouped into neighbourhoods, we used the *igraph* R package. When detecting possible communities of nodes, the detection happened with the following parameters in the *spinglass algorithm*: TODO: try the walktrap algorithm from igraph!

2.1.4. Expected Influence of nodes To understand certain items' centrality and importance, besides regularly used centrality measures (such as strength, closeness and betweenness of nodes in a network), one can also use the One- and Two-Step Expected Influence (EI) measures (**???**). Traditionally in unweighted networks (where network edges are encoded to 0 or 1) a node's centrality is the sum of edges, and in weighted nodes it is the sum of the edge weights. In this approach negative edges are treated the same way as positives, as the absolute values are used for centrality estimation. On the other hand, the Expected Influence metrics take into account if a node has positive and negative edges, too.

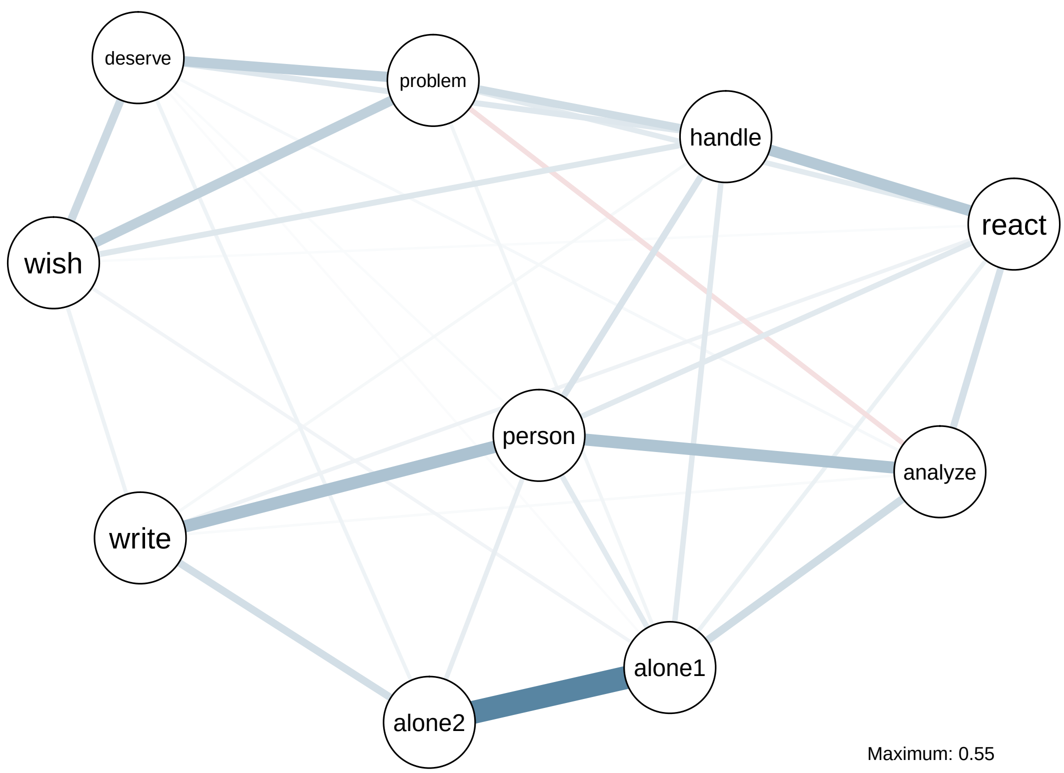
2.1.5. Identifying important nodes In order to further elaborate the question of node importance within the network, we identified *bridge nodes* using the functions provided by the *networktools* R package (Jones, 2017). We estimated both one-step and two-step bridge expected influence measures which reflect the sum of edge weights connecting one node to other groups on nodes (or communities of nodes); and also express a node's secondary influence on the rest of the network, respectively. Higher influence scores indicate that a certain node is more likely to active neighbouring nodes, subgroups of nodes or communities.

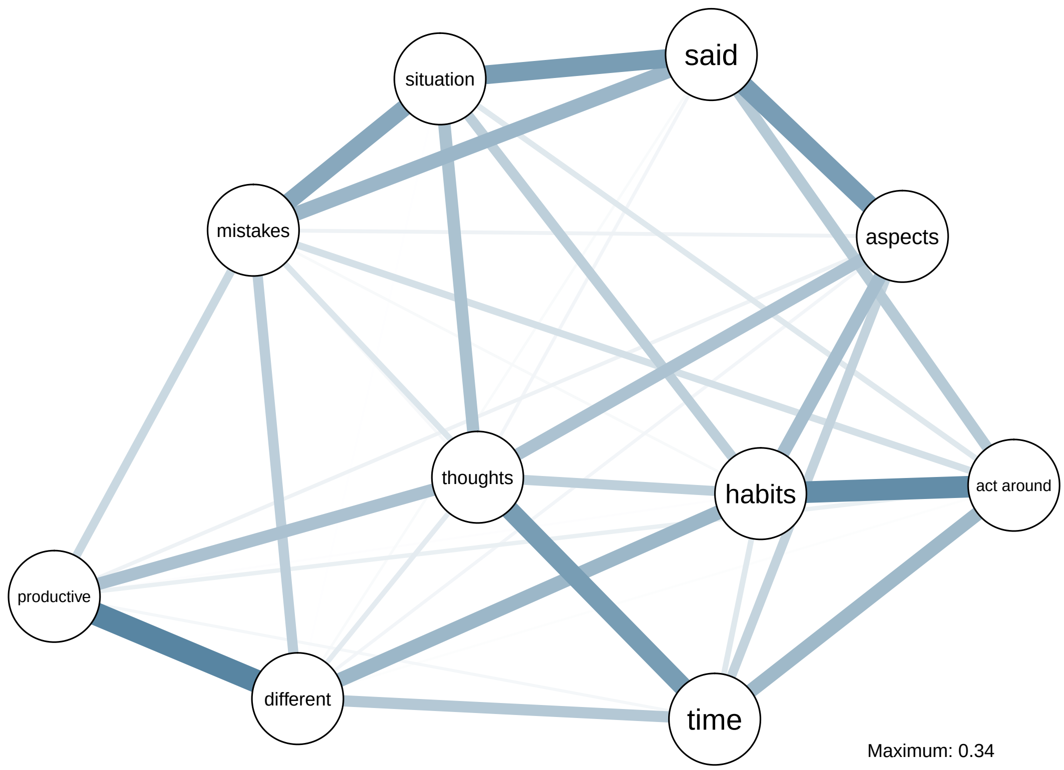
2.1.6. Network accuracy and stability Accuracy and stability have an important role in estimating network parameters when samples might vary in characteristics and size (Epskamp, Borsboom, & Fried, 2017). One way of estimating network accuracy is to produce a 95% confidence interval of edge-weights with some variant of bootstrap. In our case we chose the nonparametric variant because of interpretability. Network stability is quantified with the *Correlation Stability* (CS-coefficient): an estimate of the maximum proportion of cases that can be dropped (with 95% certainty) to retain a correlation with the original centrality of higher than 0.7. Simulation studies (Epskamp, Borsboom, & Fried, 2017) show that the CS-coefficient should be between 0.25 and 0.5.

3. Results

3.1 Network plots

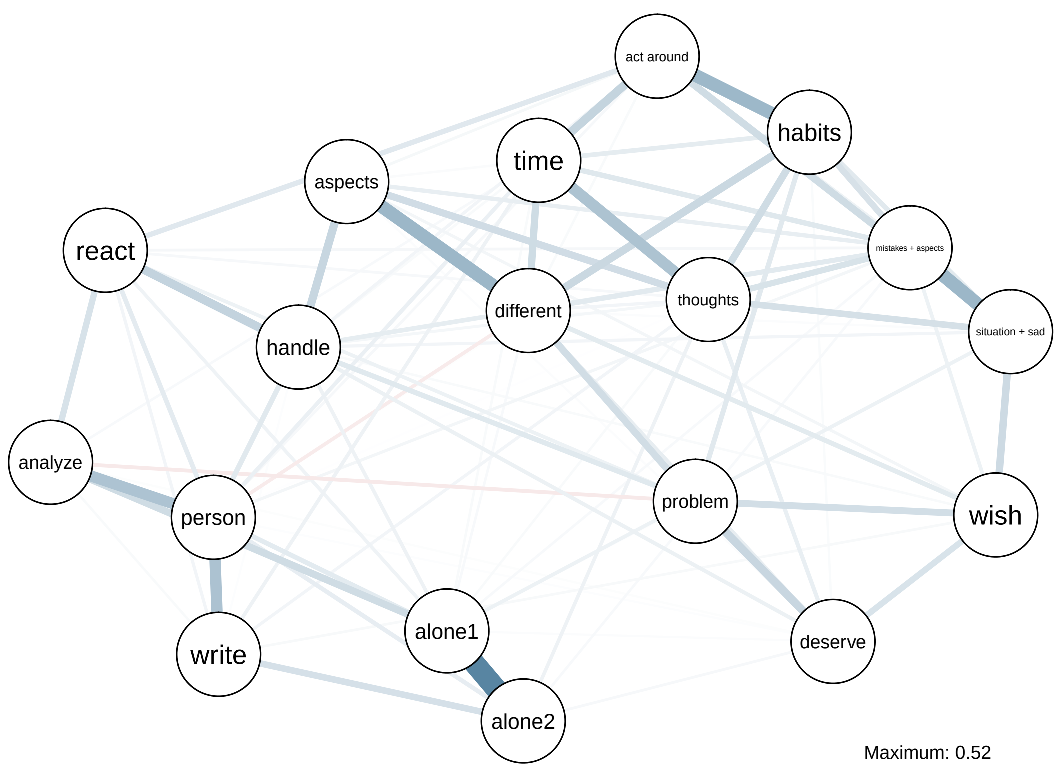
RSS network





SCRS network

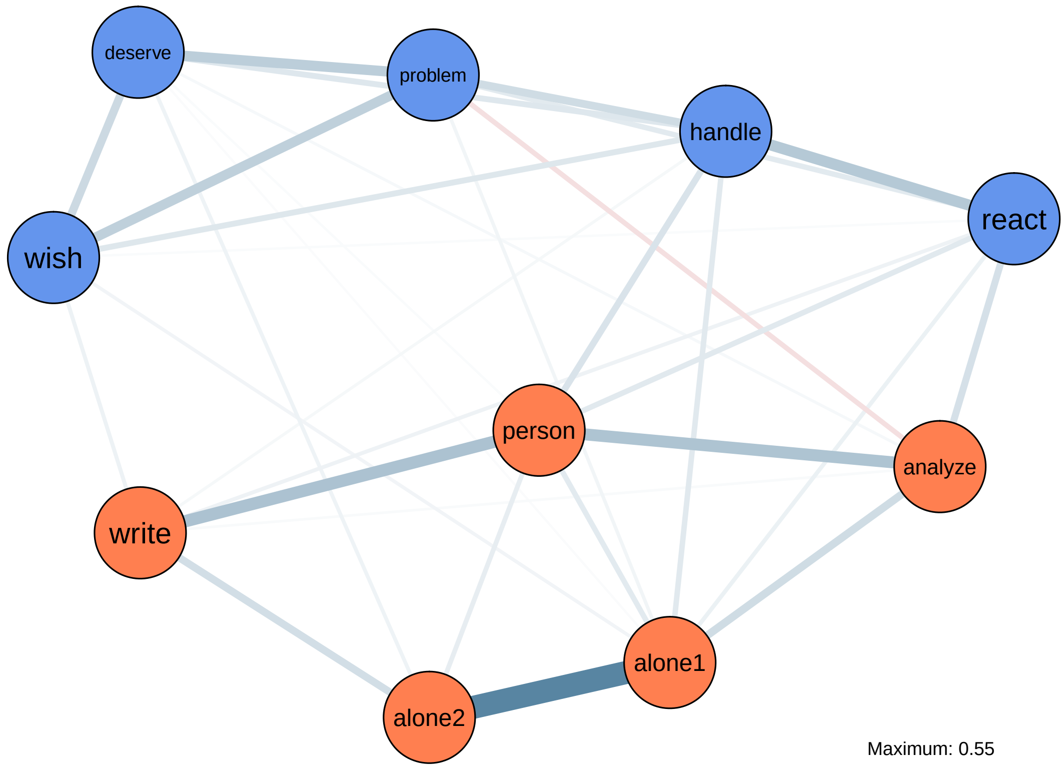
```
##    |                                                                    |
```



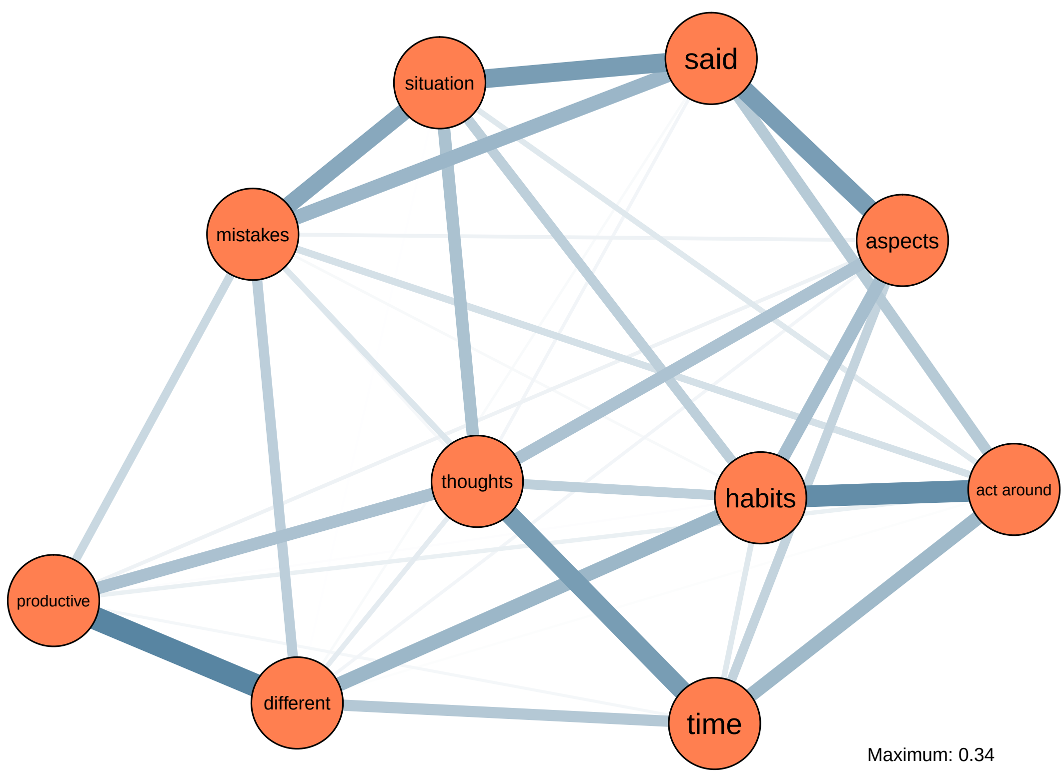
3.2 Community detection

Community mapping of RSS, MHC and SCRS with the *spinglass algorithm*:

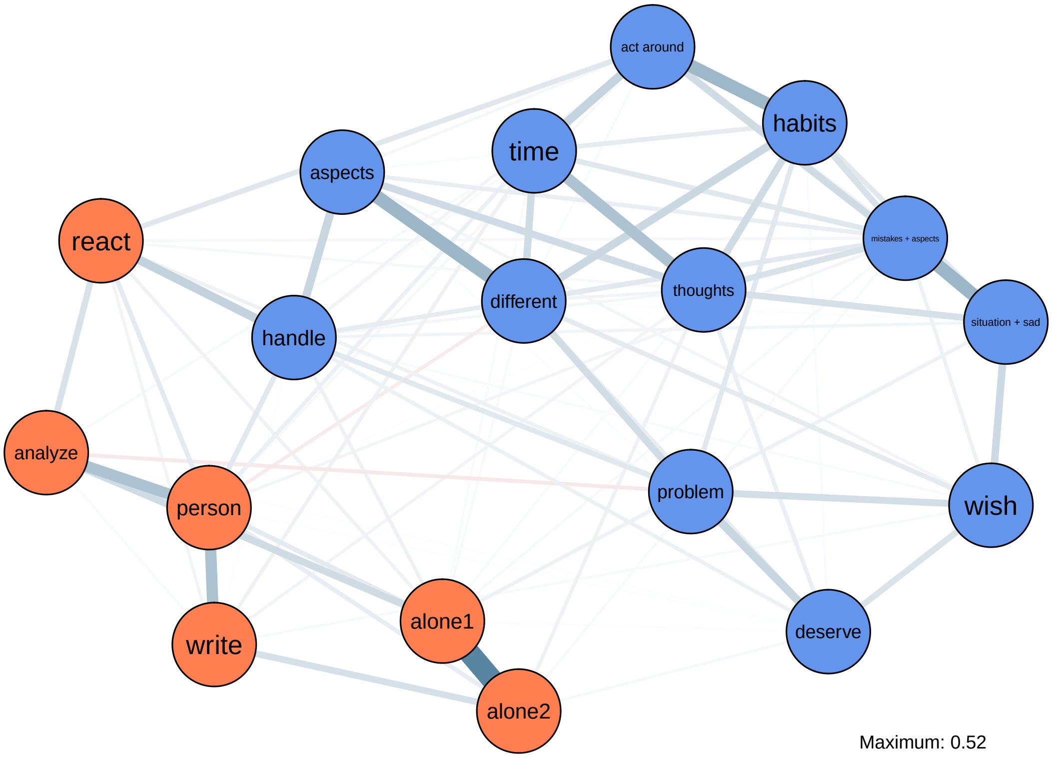
RSS communities identified with parameters *gamma = 0.5, spins = 25, starting temperature = 1, ending temperature = 0.01, cooling factor = 0.99.*



SCRS communities identified with parameters *gamma = 0.5, spins = 25, starting temperature = 1, ending temperature = 0.01, cooling factor = 0.99.*



RRS and MHC combined communities identified with parameters *gamma = 0.5, spins = 25, starting temperature = 1, ending temperature = 0.01, cooling factor = 0.99.*

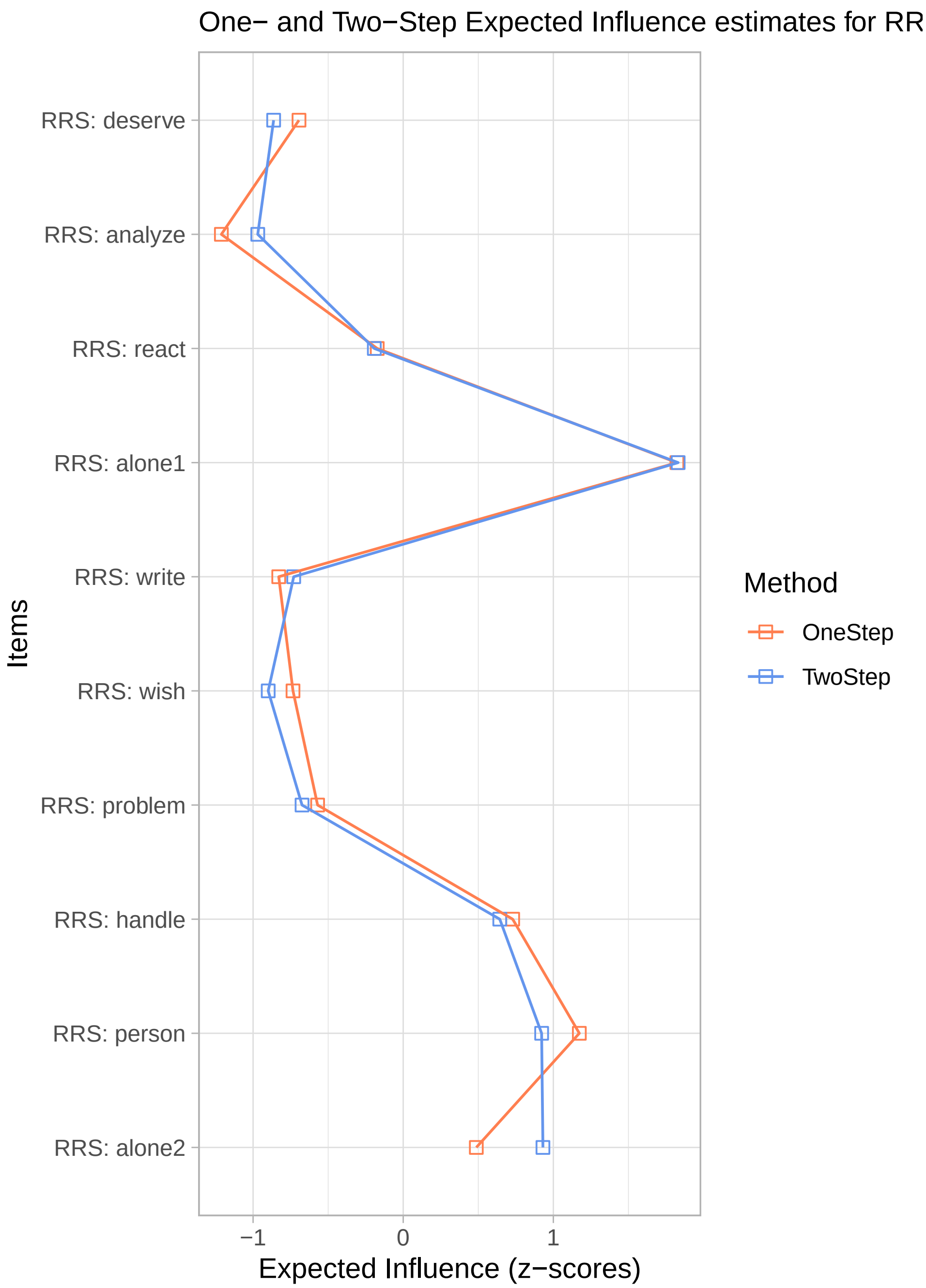


3.3 Expected Influence

In order to estimate the Expected Influence metrics, we used the *expectedInf* function of the *networktools* R package. The influence score are standardized.

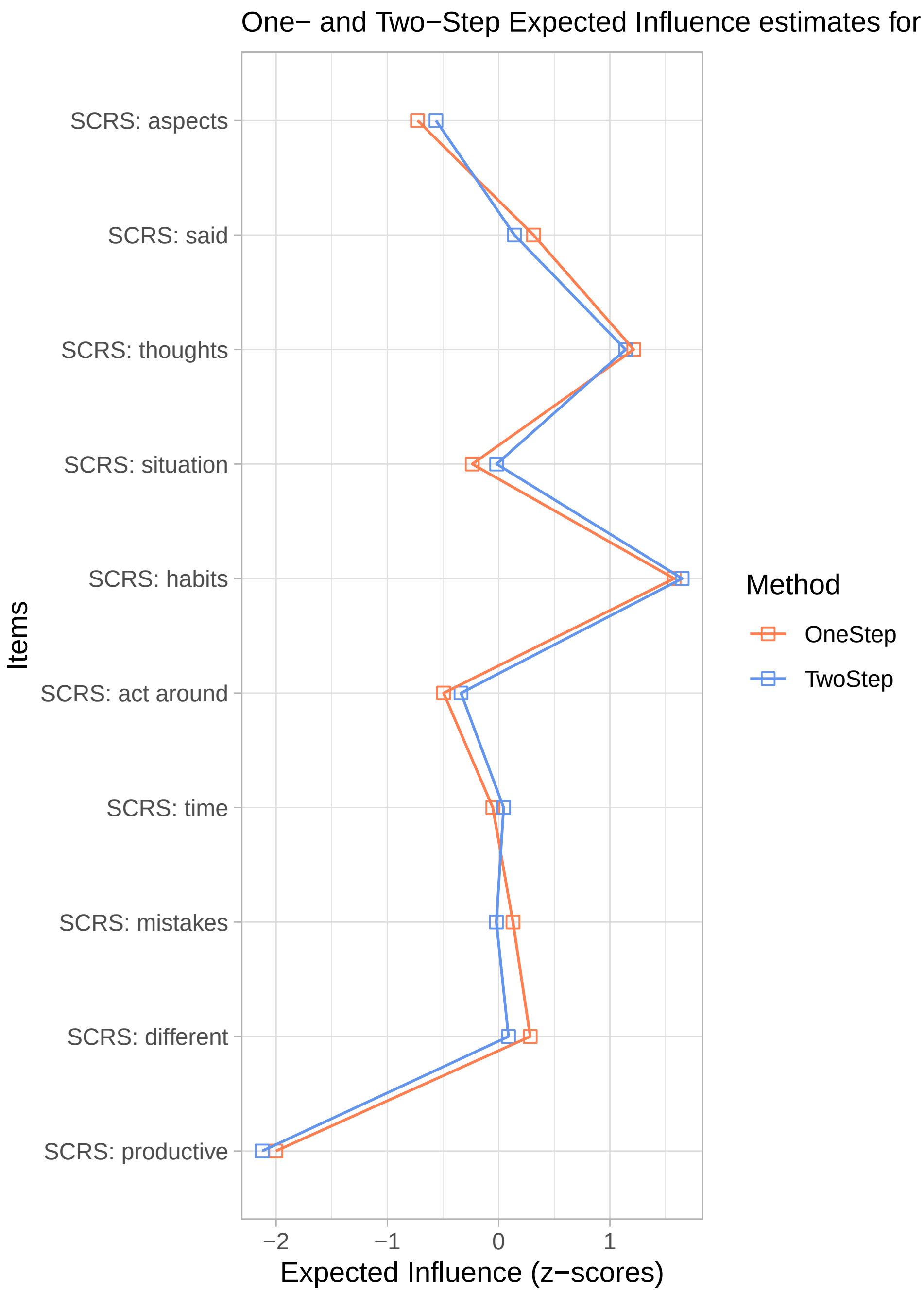
RRS Expected Influence Measures

```
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
```



SCRS Expected Influence Measures

```
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
```



Combined RRS and SCRS Expected Influence Measures

```
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
## [1] "RRS"
```

```
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
## [1] "SCRS"
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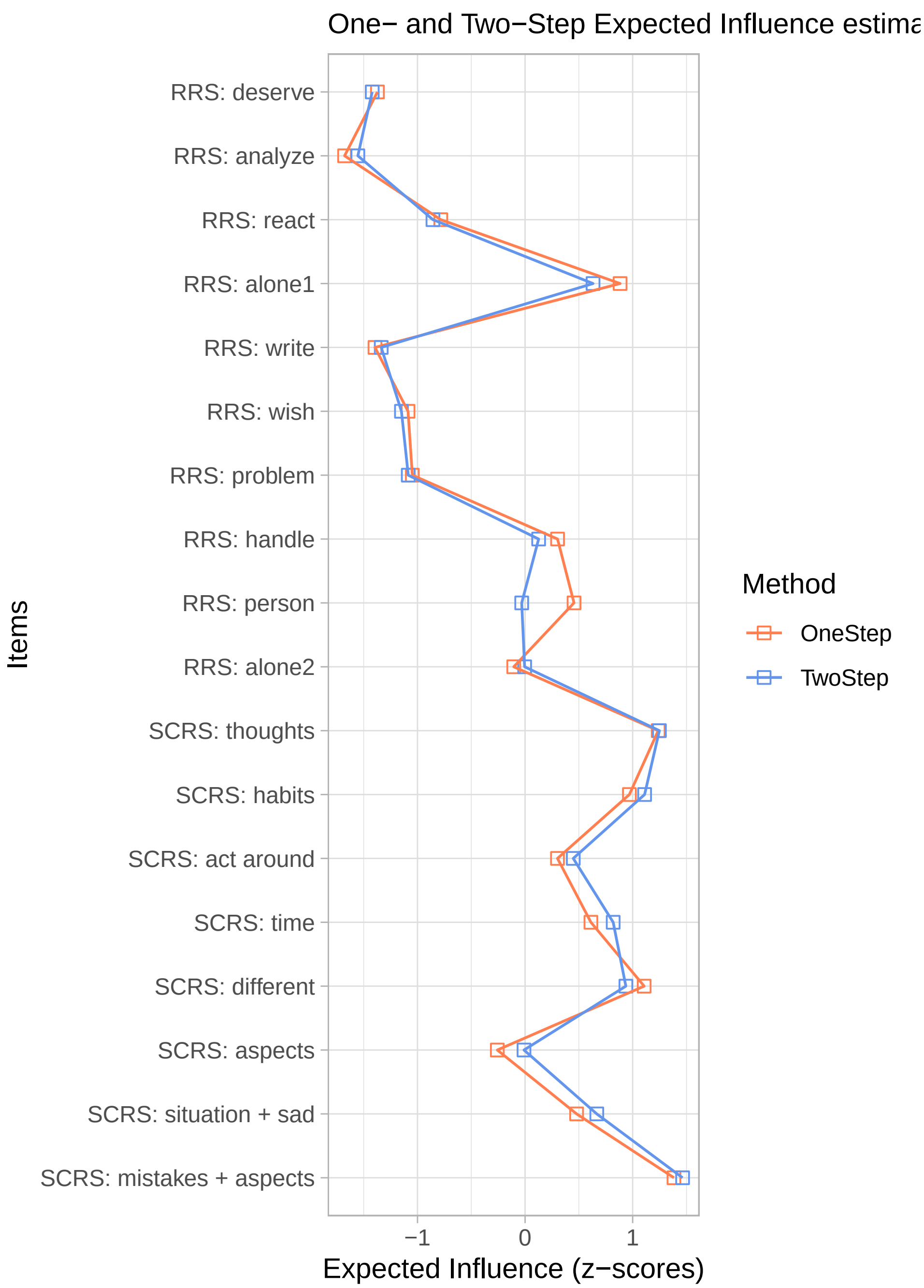
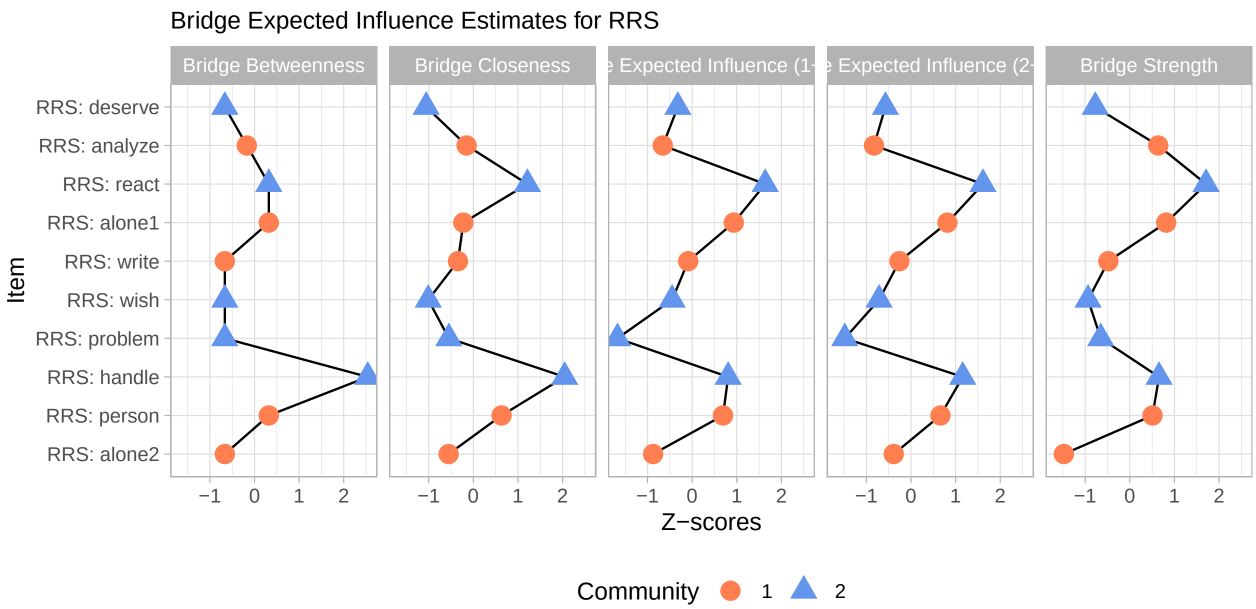


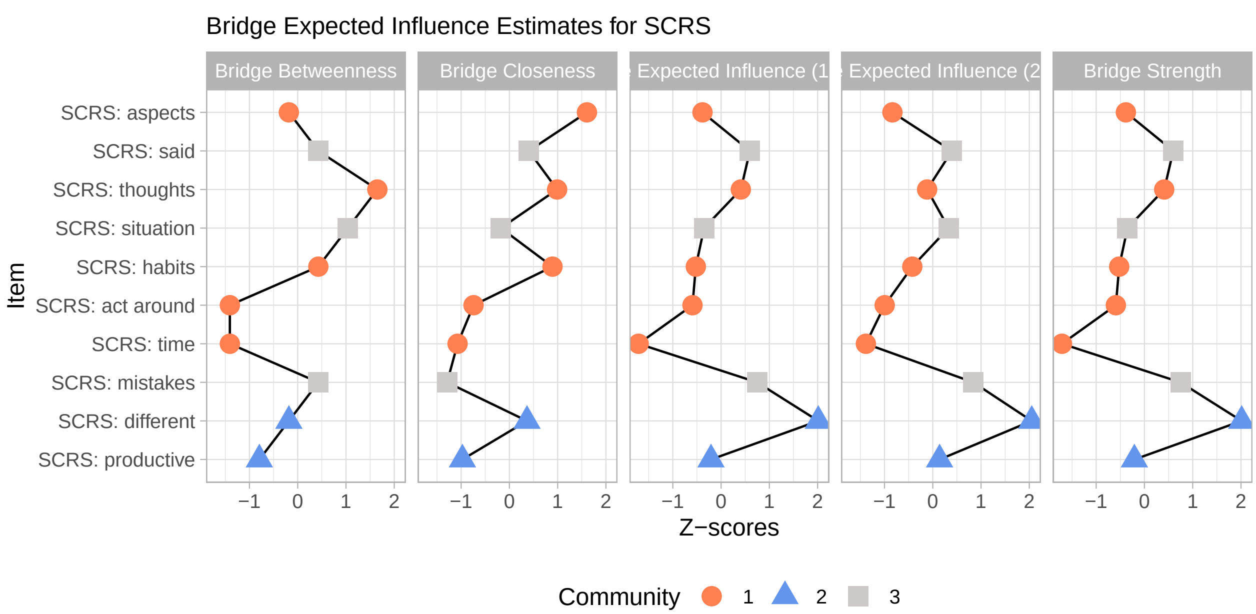
3.4 Bridge nodes

Using the *networktools* R library bridge expected influence measure were estimated for the three tools we used in this study. We also added aesthetics to reflect which items belonged to the same community according to our previous clustering steps.

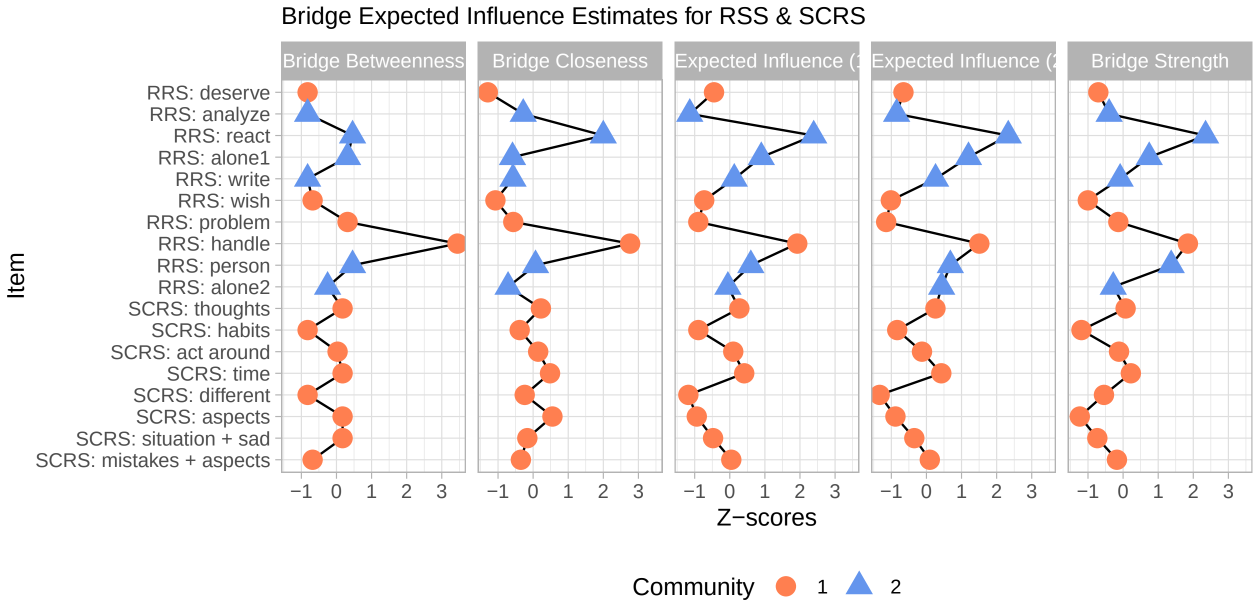
RRS bridge estimates



SCRS bridge estimates



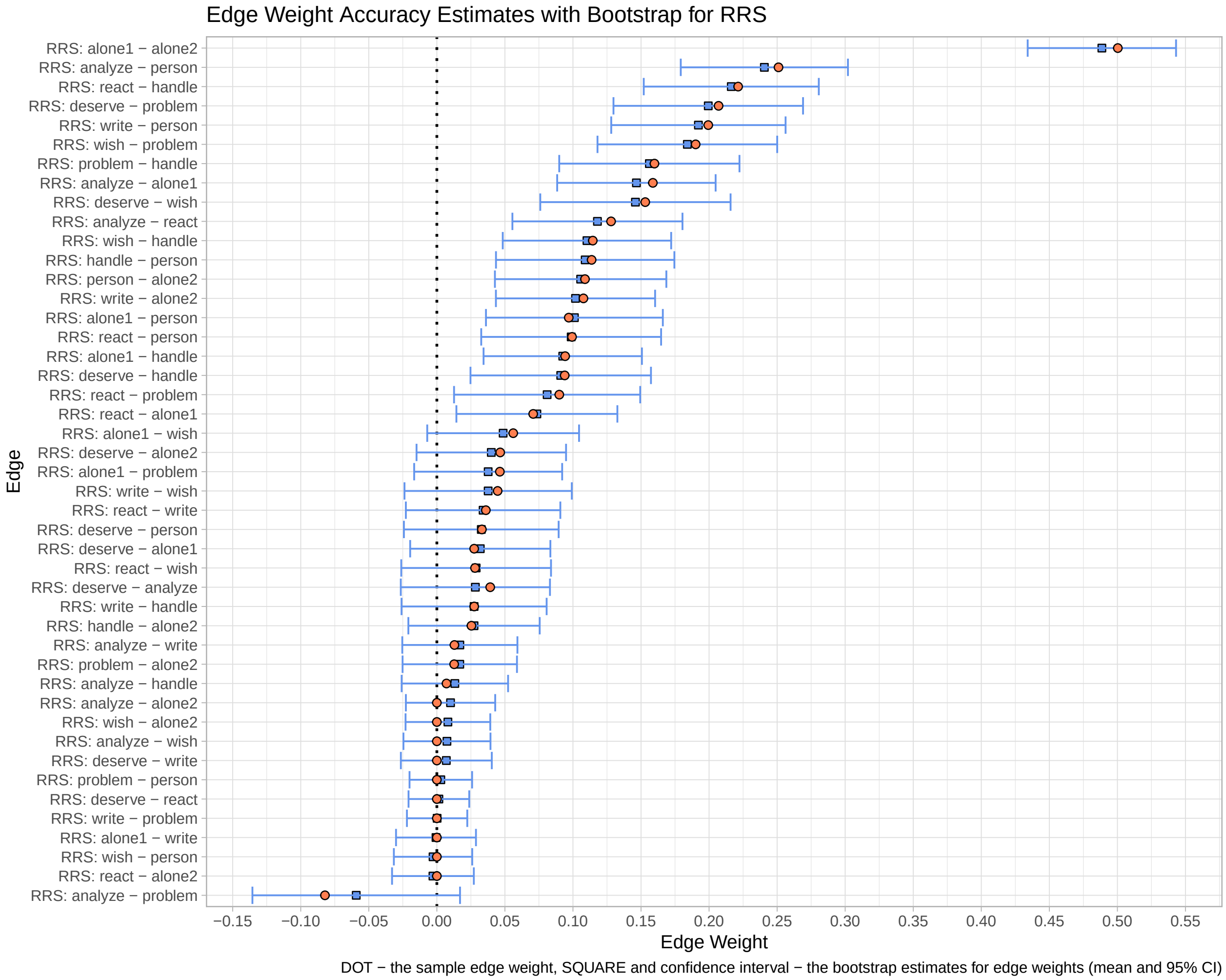
Combined RRS and SCRS bridge estimates



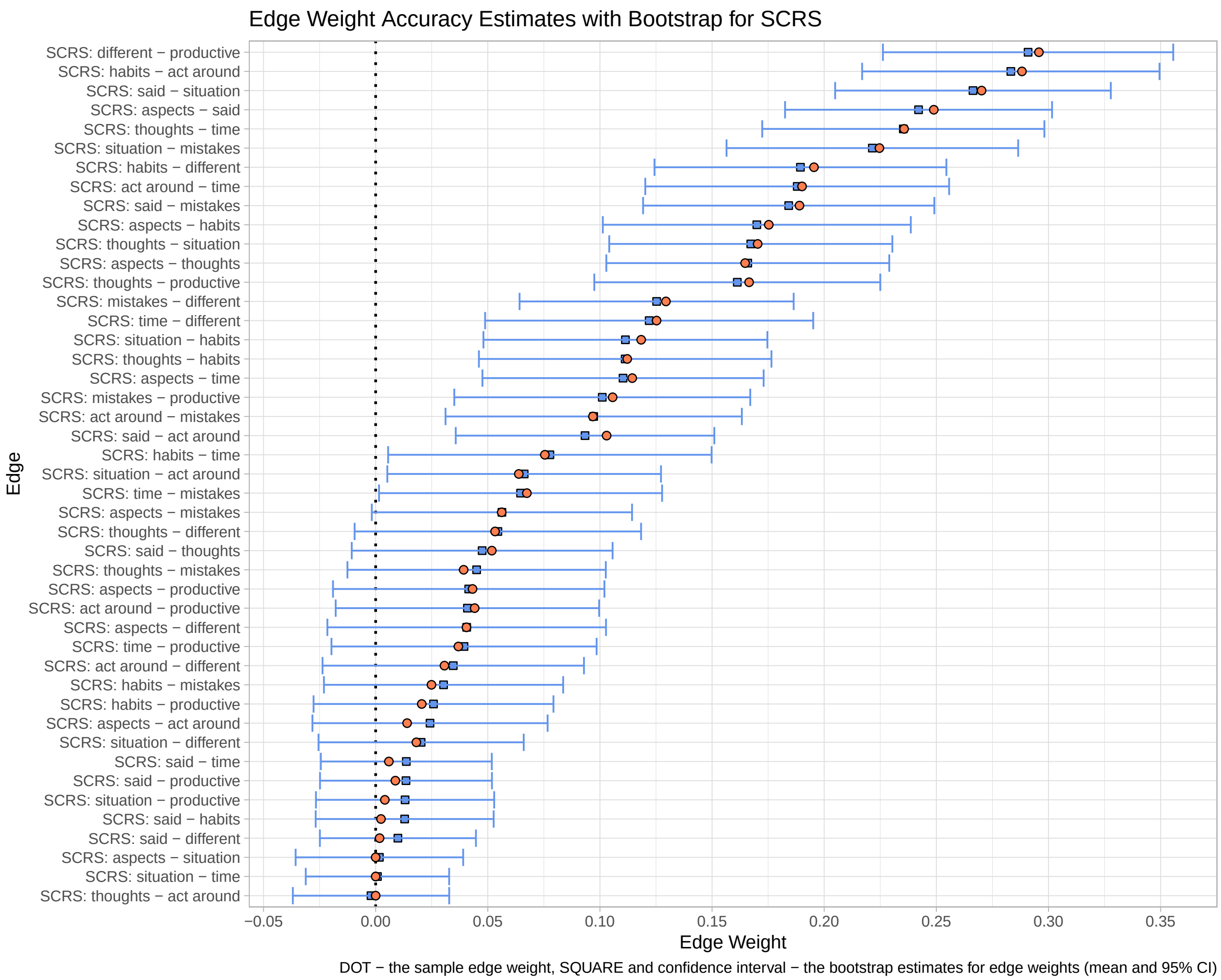
3.5 Network accuracy and stability

We used the *bootnet* R package to have an estimate of edge-weight accuracy with *nonparametric bootstrap* (1000 times sampling) and also produce measures of centrality stability with *case dropping bootstrap* (to simulate cases where only the subset of the existing data would be available).

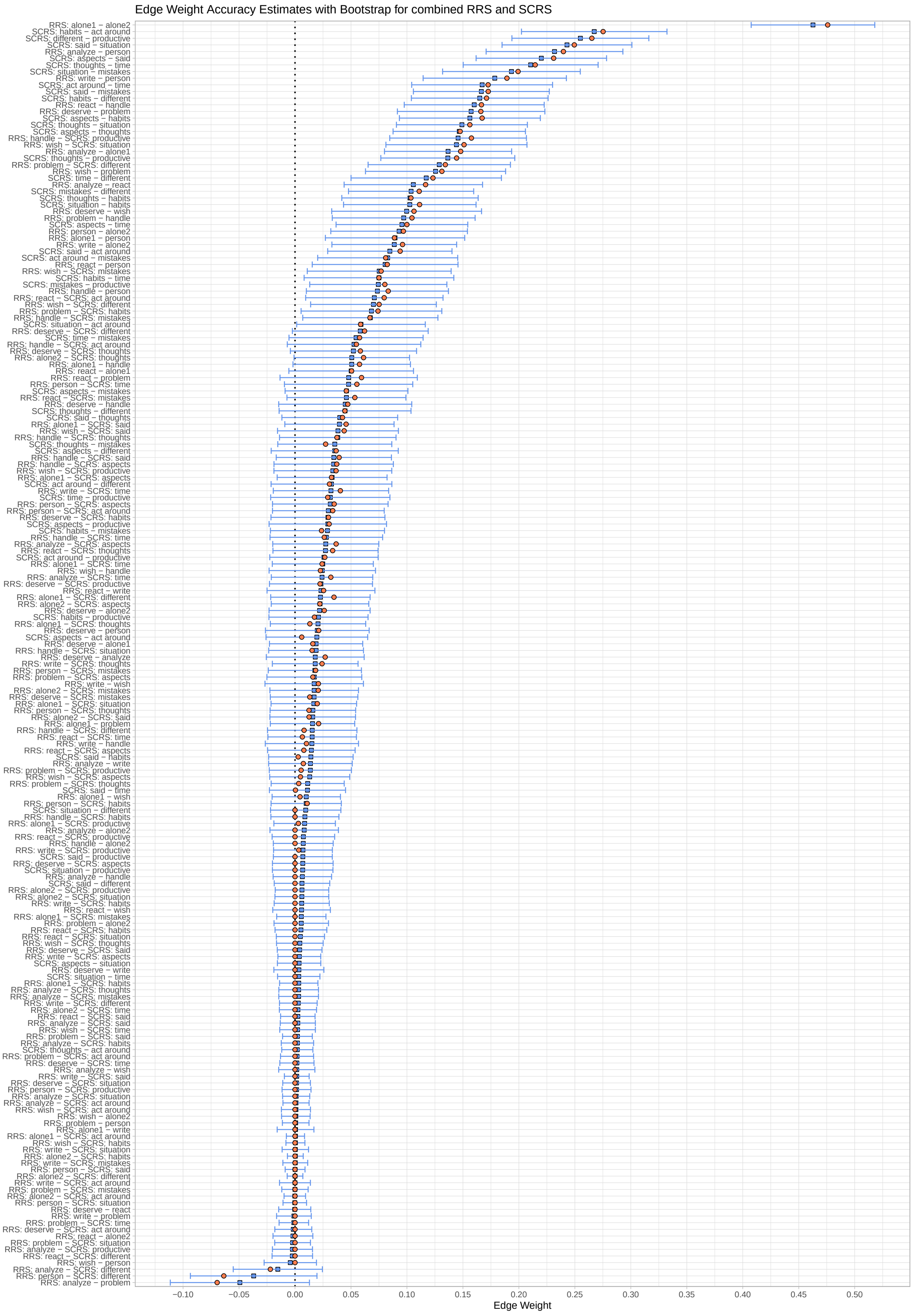
Network accuracy estimate for RRS



Network accuracy estimate for SCRS



Network accuracy estimate for combind RRS and SCRS


Edge Weight Accuracy Estimates with Bootstrap for combined RRS and SCRS
Edge
Edge Weight
−0.10 −0.05 0.00 0.05 0.10 0.15 0.20 0.25 0.30 0.35 0.40 0.45 0.50
DOT − the sample edge weight, SQUARE and confidence interval − the bootstrap estimates for edge weights (mean and 95% CI)

References

Bernstein, E. E., Heeren, A., & McNally, R. J. (2019). Reexamining trait rumination as a system of repetitive negative thoughts: A network analysis. *Journal of Behavior Therapy and Experimental Psychiatry*, *63*, 21–27. https://doi.org/10.1016/j.jbtep.2018.12.005

Epskamp, S., Borsboom, D., & Fried, E. I. (2017). Estimating psychological networks and their accuracy: A tutorial paper. *Behavior Research Methods*, *50*(1), 195–212. https://doi.org/10.3758/s13428-017-0862-1

Epskamp, S., Borsboom, D., Fried, E. I., Psychology, D. of, & Amsterdam, U. of. (2017). *Estimating psychological networks and their accuracy: Supplementary materials.*

Jones, P. (2017). *Networktools: Assorted and tools for identifying and important nodes and in networks.*

Newman, M. E. J. (2010). *Networks - an introduction.* Oxford University Press.